

Comparison of Erasure-Aware Error Correction Decoding for Fault Tolerant
Quantum Algorithms
University of California, Berkeley
C191A Introduction to Quantum Computing I
Final Project. Fall 2025

Leigh Zaman
leighzaman@berkeley.edu

Arin Sen
arin.sen@berkeley.edu>

Ilyes Tata
ilyes.tata@berkeley.edu

Last revised:
October 31, 2025

Contents

1	Introduction	1
2	Background & Related Work	1
3	Methods	1
4	Metrics	1
5	Results	1
6	Discussion	1
7	Conclusion	1
8	References	2
A	Appendix: Syndrome of Loss Errors	4
B	Appendix: Additional Figures and Tables	4

1 Introduction

Motivation & research question
Sub-questions & hypotheses

2 Background & Related Work

3 Methods

4 Metrics

5 Results

6 Discussion

7 Conclusion

8 References

- [1] Y. Wu, S. Kolkowitz, S. Puri, and J. D. Thompson, “Erasure conversion for fault-tolerant quantum computing in alkaline earth Rydberg atom arrays,” en, *Nature Communications*, vol. 13, no. 1, p. 4657, Aug. 2022, ISSN: 2041-1723. DOI: [10.1038/s41467-022-32094-6](https://doi.org/10.1038/s41467-022-32094-6). Accessed: Oct. 11, 2025. [Online]. Available: <https://www.nature.com/articles/s41467-022-32094-6>.
- [2] P. Scholl, A. L. Shaw, R. B.-S. Tsai, R. Finkelstein, J. Choi, and M. Endres, “Erasure conversion in a high-fidelity Rydberg quantum simulator,” en, *Nature*, vol. 622, no. 7982, pp. 273–278, Oct. 2023, ISSN: 0028-0836, 1476-4687. DOI: [10.1038/s41586-023-06516-4](https://doi.org/10.1038/s41586-023-06516-4). Accessed: Oct. 11, 2025. [Online]. Available: <https://www.nature.com/articles/s41586-023-06516-4>.
- [3] M. N. H. Chow, “Circuit-Based Leakage-to-Erasure Conversion in a Neutral-Atom Quantum Processor,” *PRX Quantum*, vol. 5, no. 4, 2024. DOI: [10.1103/PRXQuantum.5.040343](https://doi.org/10.1103/PRXQuantum.5.040343).
- [4] B. W. Reichardt *et al.*, *Fault-tolerant quantum computation with a neutral atom processor*, arXiv:2411.11822 [quant-ph], Jun. 2025. DOI: [10.48550/arXiv.2411.11822](https://doi.org/10.48550/arXiv.2411.11822). Accessed: Oct. 11, 2025. [Online]. Available: <http://arxiv.org/abs/2411.11822>.
- [5] L. Henriët *et al.*, “Quantum computing with neutral atoms,” en-GB, *Quantum*, vol. 4, p. 327, Sep. 2020, Publisher: Verein zur Förderung des Open Access Publizierens in den Quantenwissenschaften. DOI: [10.22331/q-2020-09-21-327](https://doi.org/10.22331/q-2020-09-21-327). Accessed: Oct. 11, 2025. [Online]. Available: <https://quantum-journal.org/papers/q-2020-09-21-327/>.
- [6] M. Saffman, *Quantum computing with atomic qubit arrays: Confronting the cost of connectivity*, arXiv:2505.11218 [quant-ph], May 2025. DOI: [10.48550/arXiv.2505.11218](https://doi.org/10.48550/arXiv.2505.11218). Accessed: Oct. 11, 2025. [Online]. Available: <http://arxiv.org/abs/2505.11218>.
- [7] M. Morgado and S. Whitlock, “Quantum simulation and computing with Rydberg-interacting qubits,” en, *AVS Quantum Science*, vol. 3, no. 2, p. 023501, Jun. 2021, ISSN: 2639-0213. DOI: [10.1116/5.0036562](https://doi.org/10.1116/5.0036562). Accessed: Oct. 17, 2025. [Online]. Available: <https://pubs.aip.org/aqs/article/3/2/023501/997323/Quantum-simulation-and-computing-with-Rydberg>.
- [8] H. Levine, “High-Fidelity Control and Entanglement of Rydberg-Atom Qubits,” *Physical Review Letters*, vol. 121, no. 12, 2018. DOI: [10.1103/PhysRevLett.121.123603](https://doi.org/10.1103/PhysRevLett.121.123603).
- [9] A. Kubica, A. Haim, Y. Vaknin, H. Levine, F. Brandão, and A. Retzker, “Erasure Qubits: Overcoming the T 1 Limit in Superconducting Circuits,” en, *Physical Review X*, vol. 13, no. 4, p. 041022, Nov. 2023, ISSN: 2160-3308. DOI: [10.1103/PhysRevX.13.041022](https://doi.org/10.1103/PhysRevX.13.041022). Accessed: Oct. 18, 2025. [Online]. Available: <https://link.aps.org/doi/10.1103/PhysRevX.13.041022>.
- [10] J. Thompson, *Erasure conversion for fault-tolerant quantum computing in alkaline earth Rydberg atom arrays*, en, Jun. 2022. DOI: [10.7910/DVN/H9LV4H](https://doi.org/10.7910/DVN/H9LV4H). Accessed: Oct. 18, 2025. [Online]. Available: <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/H9LV4H>.
- [11] C. Gidney, *Stim: A fast stabilizer circuit simulator*, Pages: 497 Publication Title: Quantum Volume: 5 original-date: 2021-02-19T15:55:47Z, Jul. 2021. Accessed: Oct. 19, 2025. [Online]. Available: <https://doi.org/10.22331/q-2021-07-06-497>.
- [12] oscarhiggott, *Oscarhiggott/PyMatching*, original-date: 2019-11-08T11:05:08Z, Oct. 2025. Accessed: Oct. 19, 2025. [Online]. Available: <https://github.com/oscarhiggott/PyMatching>.
- [13] H. Silvério *et al.*, “Pulser: An open-source package for the design of pulse sequences in programmable neutral-atom arrays,” en-GB, *Quantum*, vol. 6, p. 629, Jan. 2022, Publisher: Verein zur Förderung des Open Access Publizierens in den Quantenwissenschaften. DOI: [10.22331/q-2022-01-24-629](https://doi.org/10.22331/q-2022-01-24-629). Accessed: Oct. 19, 2025. [Online]. Available: <https://quantum-journal.org/papers/q-2022-01-24-629/>.
- [14] D. Bluvstein, S. Evered, A. A. Geim, and a. et al., “Logical quantum processor based on reconfigurable atom arrays,” *Nature*, no. 626, pp. 58–65, 2024. DOI: <https://doi.org/10.1038/s41586-023-06927-3>.
- [15] G. Baranes *et al.*, *Leveraging Atom Loss Errors in Fault Tolerant Quantum Algorithms*, arXiv:2502.20558 [quant-ph] version: 1, Feb. 2025. DOI: [10.48550/arXiv.2502.20558](https://doi.org/10.48550/arXiv.2502.20558). Accessed: Nov. 1, 2025. [Online]. Available: <http://arxiv.org/abs/2502.20558>.

- [16] K. Sahay, “High-Threshold Codes for Neutral-Atom Qubits with Biased Erasure Errors,” *Physical Review X*, vol. 13, no. 4, 2023. DOI: [10.1103/PhysRevX.13.041013](https://doi.org/10.1103/PhysRevX.13.041013).
- [17] B. Zhang *et al.*, *Leveraging erasure errors in logical qubits with metastable ^{171}Yb atoms*, arXiv:2506.13724 [quant-ph], Jun. 2025. DOI: [10.48550/arXiv.2506.13724](https://doi.org/10.48550/arXiv.2506.13724). Accessed: Nov. 1, 2025. [Online]. Available: <http://arxiv.org/abs/2506.13724>.
- [18] K. Sahay, J. Jin, J. Claes, J. D. Thompson, and S. Puri, “High-Threshold Codes for Neutral-Atom Qubits with Biased Erasure Errors,” *Physical Review X*, vol. 13, no. 4, p. 041 013, Oct. 2023, Publisher: American Physical Society. DOI: [10.1103/PhysRevX.13.041013](https://doi.org/10.1103/PhysRevX.13.041013). Accessed: Nov. 1, 2025. [Online]. Available: <https://link.aps.org/doi/10.1103/PhysRevX.13.041013>.
- [19] S. Ma *et al.*, “High-fidelity gates and mid-circuit erasure conversion in an atomic qubit,” en, *Nature*, vol. 622, no. 7982, pp. 279–284, Oct. 2023, Publisher: Nature Publishing Group, ISSN: 1476-4687. DOI: [10.1038/s41586-023-06438-1](https://doi.org/10.1038/s41586-023-06438-1). Accessed: Nov. 1, 2025. [Online]. Available: <https://www.nature.com/articles/s41586-023-06438-1>.
- [20] Y. Li, Y. Bao, M. Peper, C. Li, and J. D. Thompson, *Fast, continuous and coherent atom replacement in a neutral atom qubit array*, arXiv:2506.15633 [quant-ph], Jun. 2025. DOI: [10.48550/arXiv.2506.15633](https://doi.org/10.48550/arXiv.2506.15633). Accessed: Nov. 1, 2025. [Online]. Available: <http://arxiv.org/abs/2506.15633>.
- [21] M. A. Norcia *et al.*, “Midcircuit Qubit Measurement and Rearrangement in a Yb 171 Atomic Array,” en, *Physical Review X*, vol. 13, no. 4, p. 041 034, Nov. 2023, ISSN: 2160-3308. DOI: [10.1103/PhysRevX.13.041034](https://doi.org/10.1103/PhysRevX.13.041034). Accessed: Nov. 1, 2025. [Online]. Available: <https://link.aps.org/doi/10.1103/PhysRevX.13.041034>.

A Appendix: Syndrome of Loss Errors

Describe how a loss is heralded during state readout. Each potential loss location is described by a distinct error probability, described by a unique configuration of the error syndroms. [EXAMPLE OF ERROR SYNDROME]

B Appendix: Additional Figures and Tables